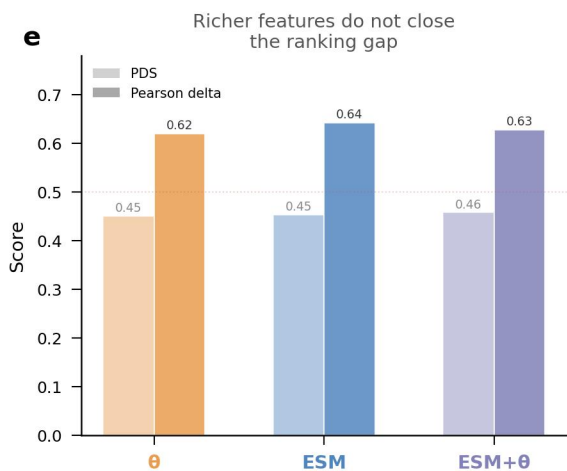
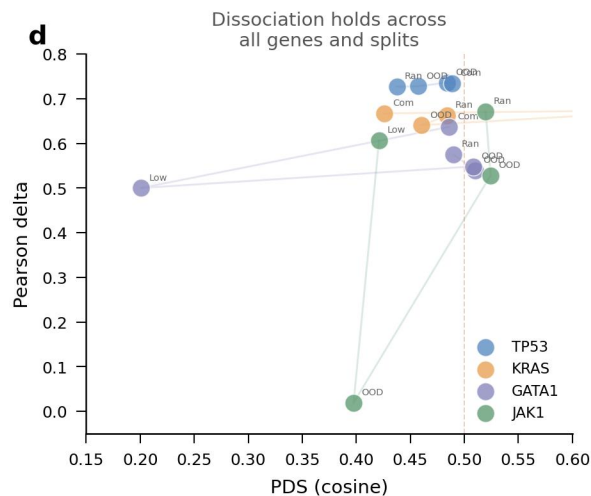
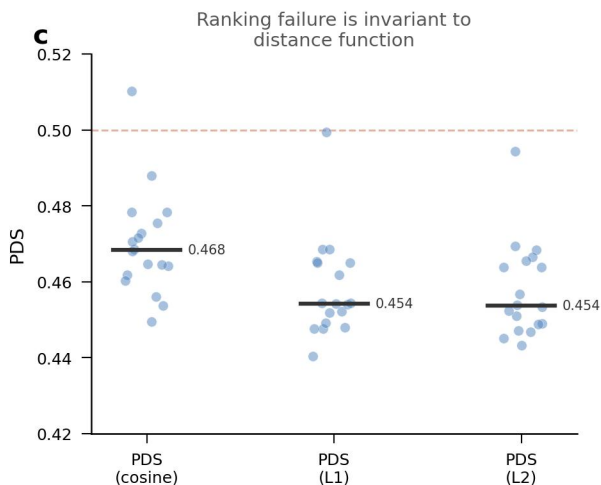
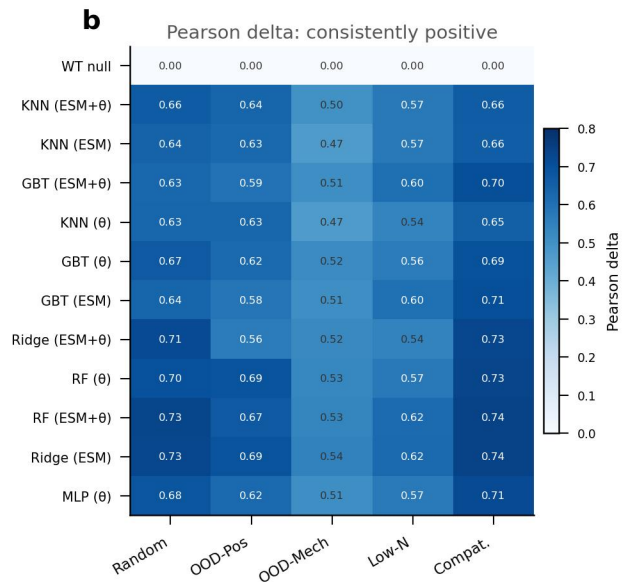
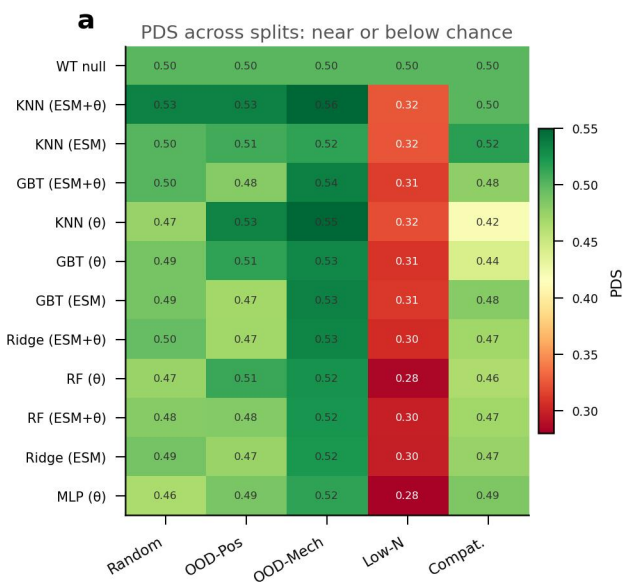




f Existing methods assume gene-level perturbation identity

Method	Failure mode	Outcome
CPA	Dose scalar input	NaN on θ vector
CellFlow	No unseen-variant API	Train-set only
STATE	Per-dataset retrain	GPU OOM (GATA1)
GEARS	Gene-keyed graph	Allele-blind
scDFM	No public checkpoint	Cannot reproduce
Biolord	Ordered attributes	PDS $\approx$ chance